Academic analysis of Indian digital payment fraud

"What academic studies analyse UPI or digital payment fraud in India, including mule accounts, victim profiles and detection methods?" *Generated by LeapSpace*

Direct answer

The academic literature on **UPI and digital payment fraud in India** clusters around three main areas: **fraud-detection methods**, **mule-account detection**, and **victim/user vulnerability profiles**. The strongest body of work is on detection models, while mule-account research is much narrower and appears in only a few studies. [30] [136] [138] [141] [148] [155] [156] [157] [158] [159] [160] [161] [162] [163] [164] [165] [166] [167] [168] [169] [170] [171]

What the studies analyse

Theme	Studies / focus	Main contribution
Fraud detection methods	UPI fraud detection with ML, deep learning, ensembles, anomaly detection, and explainable AI [136] [138] [141] [148] [158] [159] [160] [161] [162] [163] [164] [165] [166] [167] [171]	These papers model fraudulent vs genuine UPI transactions using transaction behavior, device, time, location, frequency, and related features. [136] [138] [141] [148] [158] [159] [160] [161] [162] [163] [164] [165] [166] [167] [171]
Mule accounts	Money mule detection in UPI and broader instant-payment gateways [30] [155]	**MuleTrack** is explicitly about **money mule accounts in UPI** and uses domain heuristics plus a Markov-chain temporal model; the gateway paper also flags mule networks as a major threat and proposes recipient validation plus adaptive risk scoring. [30] [155]
Victim profiles / user vulnerability	Phishing/vishing, trust, awareness, and demographic or regional patterns [156] [157] [168] [169] [170]	These studies show that user awareness, trust, perceived risk, security assurances, and regional incident patterns shape vulnerability to UPI-related fraud and scams. [156] [157] [168] [169] [170]
Operational prevention	Real-time scoring, adaptive alerts, step-up authentication, and anomaly-aware gating [138] [148] [155] [162]	Rather than simple blocking, several studies emphasize graduated intervention to reduce false positives while protecting legitimate users. [138] [148] [155] [162]

Most directly relevant studies

Mule-account focused

- **MuleTrack: A Lightweight Temporal Learning Framework for Money Mule Detection in Digital Payments** is the clearest mule-specific study, and it explicitly targets **money mule accounts in UPI** using temporal behavior modeling. [30]
- **Instant Fraud Detection Gateway for Secure Digital Transactions** includes mule networks among the main digital-payment threats and proposes preventive validation and risk scoring. [155]

Victim profiles and scam exposure

- **Unified Payments Interface (UPI): Fraud and Prevention Strategies** discusses fraud tactics and user vulnerability, including lack of awareness about fraudster methods. [156]
- **The Rise of Phishing and Vishing in Digital Payments: An Analytical Review of NCRB Data (2016–2025)** emphasizes rising incidents, regional hotspots, and gaps in user knowledge and enforcement. [157]
- **Unpacking Trust in UPI Micro-Payments: A Techno-Behavioural Perspective on Gen Z Users** studies how trust is shaped by security measures, regulatory assurance, speed, reliability, ease of use, data transparency, and past experience. [168]
- **Cashless preferences during the COVID-19 pandemic: investigating user intentions to continue UPI-based payment systems in India** also addresses user intentions and trust-related adoption factors. [169]
- **A fractional factorial design for evaluating digital wallet attribute utilities: evidence from Indian E-payment systems** contributes evidence on which digital-wallet attributes users value. [170]

Detection-method papers

A large set of papers focus on machine-learning-based fraud detection for UPI, including:

- **Machine Learning for Safeguarding of UPI Scams** [136]
- **Enhancing Digital Payment Security: UPI Fraud Detection with Advanced Machine Learning Algorithms** [138]
- **UPI Fraud Detection using Machine Learning** [141]
- **Smart UPI Fraud Detect: Ml-Based Anomaly Identification in UPI Payments** [148]
- **Comparative Analysis of UPI Fraud Detection using Machine Learning** [158]
- **Detecting UPI Fraud Using Machine Learning Techniques in Python** [159]
- **Unified Payment Interface Imposture and Scam Detection Using Deep Learning And ANN** [160]
- **Enhancing UPI Security with AI: An Ensemble-Based Fraud Detection Model** [161]
- **UPI Sentinel: Anomaly-Aware Fraud Detection Using Hybrid ML Models** [162]

- **Trust Pay UPI: A Machine Learning Approach for Detecting Fraud in UPI Transactions** [163]
- **Secure Digital Payments: Fraud Detection in UPI Using Machine Learning Techniques** [164]
- **AI-Driven Real-Time Threat Detection for UPI Transactions** [165]
- **A Novice Approach for UPI Fraud Detection by Using Machine Learning Techniques** [166]
- **Enhanced Digital Payment Security Using Predictive Analytics and AI-Powered Anomaly Detection Techniques** [167]
- **Experimental Evaluation Unified Payment Interface (UPI) Fraud Detection System Using Elevated Deep Learning Methodology** [171]

Bottom line

If you want the **best evidence on methods**, the literature is rich and mostly centered on ML, deep learning, anomaly detection, and ensemble systems for transaction-level fraud detection. [136] [138] [141] [148] [158] [159] [160] [161] [162] [163] [164] [165] [166] [167] [171] If you want **mule accounts specifically**, the evidence is limited but direct, especially in MuleTrack. [30] [155] If you want **victim profiles**, the literature focuses more on trust, awareness, phishing/vishing exposure, and adoption behavior than on a single unified victim typology. [156] [157] [168] [169] [170]

Confidence level: High — the cited studies consistently support the three-way split between detection methods, mule-account analysis, and victim/user vulnerability, with especially strong coverage of detection methods and limited but direct coverage of mule accounts.

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